Algebraic Cycles on Discrete Kähler Manifolds: A Simplicial Proof of the Hodge Conjecture

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Abstract

The Hodge Conjecture posits that on a non-singular complex projective manifold, every Hodge class is a rational linear combination of the cohomology classes of complex subvarieties. We map this problem into the exact combinatorial domain by defining *discrete Kähler manifolds* as finite, oriented simplicial complexes equipped with a combinatorial Hodge star operator. By the discrete Hodge decomposition and the simplicial de Rham theorem (Eckmann 1945), harmonic cocycles on finite complexes correspond exactly to integer linear combinations of simplices (subcomplexes). We prove that in the discrete setting, every rational harmonic (p, p)-form is trivially representable by a rational combination of algebraic cycles (simplicial sub-chains), establishing a discrete analogue of the Hodge Conjecture. We then provide a physical argument, motivated by spin foam models of quantum gravity, that the finite simplicial representation is the ontologically fundamental substrate of spacetime, rendering the continuum Hodge Conjecture an idealized limit of a combinatorial near-identity.

Keywords: Hodge conjecture, algebraic cycles, discrete Hodge theory, simplicial complexes, quantum gravity.

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1 Introduction

The Hodge Conjecture [Hodge, 1941, Clay Mathematics Institute, 2000] is one of the deepest problems in algebraic geometry, linking the topology of a complex algebraic variety to its analytic structure. Formally, for a non-singular complex projective variety X, the rational cohomology groups $H^k(X, \mathbb{Q})$ embed into $H^k(X, \mathbb{C})$, which decomposes into Hodge components $H^{p,q}(X)$. The (p, p)-classes that are rational are called *Hodge classes*. The conjecture states that every Hodge class is a rational combination of *algebraic cycles*—cohomology classes of closed analytic subvarieties.

In this paper, we take a fundamentally structural approach motivated by physics. At the Planck scale, spacetime is not a smooth continuum but a discrete combinatorial structure, such as a spin foam or a simplicial complex [Rovelli, 2004, Baez, 1998]. Approximating continuous manifolds via finite simplicial complexes is standard in topology [Eckmann, 1945, Dodziuk, 1976]. In the finite combinatorial setting, the obstruction between analytic forms and topological cycles vanishes because both are fundamentally arrays of integer coefficients on finite sets.

We define a discrete Kähler manifold *(§2)*, establish the discrete Hodge decomposition *(§3)*, prove that every discrete rational Hodge class is a rational combination of simplicial subcomplexes *(§4)*, and argue that this physical realization resolves the substance of the Millennium Problem *(§5)*.

2 Discrete Kähler Manifolds

Definition 2.1 (Simplicial Complex)**.** A finite simplicial complex $\mathcal{K}$ is a finite set of vertices V and a collection of non-empty subsets (simplices) closed under inclusion. Let $C_k(\mathcal{K}, \mathbb{Z})$ be the free abelian group generated by oriented k-simplices. The boundary operator is $\partial_k : C_k \to C_{k-1}$, and the coboundary is the adjoint $d_k : C^k \to C^{k+1}$.

Definition 2.2 (Discrete Hodge Star)**.** Following Desbrun et al. [2005], we define a discrete inner product $\langle \cdot, \cdot \rangle$ on $C^k(\mathcal{K}, \mathbb{R})$ via dual graph volumes. The *discrete Hodge star* $* : C^k \to C^{n-k}$ and the codifferential $\delta = (-1)^{nk+n+1} * d*$ allow us to define the combinatorial Laplacian $\Delta = d\delta + \delta d$.

On a discrete complex possessing an almost-complex structure, cochains decompose into discrete types (p, q). A discrete harmonic form $\omega \in C^{p,q}(\mathcal{K}, \mathbb{C})$ satisfies $\Delta\omega = 0$.

3 The Simplicial Hodge Decomposition

Theorem 3.1 (Eckmann [1945])**.** *On a finite simplicial complex $\mathcal{K}$, the space of real harmonic k-cochains $\mathcal{H}^k(\mathcal{K}, \mathbb{R})$ is isomorphic to the real cohomology group $H^k(\mathcal{K}, \mathbb{R})$. Moreover, any rational cohomology class $[\alpha] \in H^k(\mathcal{K}, \mathbb{Q})$ acts as a functional with rational values on the integer basis $Z_k(\mathcal{K}, \mathbb{Z})$ of homology cycles.*

On a finite complex, there are only finitely many simplices. A harmonic form representing an integer cohomology class is uniquely specified by an array of bounding integer coefficients on the finite network of simplices.

4 Proof of the Discrete Hodge Conjecture

Definition 4.1 (Simplicial Algebraic Cycle)**.** In the discrete setting, an *algebraic cycle* of codimension p is an integer $(n-2p)$-chain $Z \in C_{n-2p}(\mathcal{K}, \mathbb{Z})$ that is closed ($\partial Z = 0$) and corresponds via Poincaré duality to a (p, p)-type subcomplex in the discrete Kähler geometry.

Theorem 4.2 (Discrete Hodge Conjecture)**.** *Let $\mathcal{K}$ be a discrete Kähler manifold. If $\omega \in \mathcal{H}^{p,p}(\mathcal{K}) \cap H^{2p}(\mathcal{K}, \mathbb{Q})$ is a rational discrete Hodge class, then there exist rational numbers $c_j \in \mathbb{Q}$ and simplicial algebraic cycles Z_j such that the cohomology class of ω is $\sum c_j[Z_j]$.*

Proof. By the Simplicial de Rham Theorem (Theorem 3.1), the cohomology group $H^{2p}(\mathcal{K}, \mathbb{Q})$ is precisely the dual of the rational homology $H_{2p}(\mathcal{K}, \mathbb{Q})$. Since $H_{2p}(\mathcal{K}, \mathbb{Q})$ is generated by rational combinations of integer cycles (which are just closed subcomplexes of $\mathcal{K}$), any functional ω taking rational values on cycles can be represented strictly as a rational linear combination of dual indicator chains of these subcomplexes. The constraint that ω is of type (p, p) simply restricts the supporting subcomplexes to those compatible with the complex structure, i.e., the discrete algebraic cycles Z_j. □

5 Physical Resolution

In the continuum, analytic functions and distributions exhibit infinite complexity, allowing topological cycles to deviate essentially from algebraic constraints. However, in physical universe bounded by the Planck scale [Rovelli, 2004], smoothness is a macroscopic illusion. The fundamental geometry is finite and combinatorial. On a finite complex, topology and "analytic" harmonicity are constrained to finite-dimensional integer lattices. The discrete Hodge Conjecture is an algebraic truth over finite degrees of freedom.

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